

Mapping and Partial Characterization of Proteases Expressed by a CHO Production Cell Line

H. Sandberg,¹ D. Lütkemeyer,² S. Kuprin,¹ M. Wrangel,¹ A. Almstedt,¹
P. Persson,¹ V. Ek,¹ M. Mikaelsson³

¹Department of Biopharmaceuticals, Biovitrum AB, SE-112 76 Stockholm, Sweden; telephone: +4686972568; fax: +4686972339; e-mail: maria.wrangel@biovitrum.com

²Cell Culture Technology, Faculty of Technology, University of Bielefeld, Bielefeld, Germany

³Kabio HB, Akersberga, Sweden

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Abstract: The proteolytic activities expressed by a Chinese hamster ovary (CHO) cell line in the cultivation supernatant during the production of recombinant factor VIII were mapped with a broad spectrum protease assay and a series of different types of protease inhibitors. The destabilizing effect on the product emanated from a metalloproteinase, which could be effectively blocked by chelating agents to lead to product stabilization. Amino acid sequences of the isolated metalloproteinase were found to have sequence homology with matrix metalloproteinases (MMPs) MMP3, MMP10, and MMP12. Several species with metalloproteinase activity were characterized and found to be related to each other. The results indicate that an MMP pro-enzyme of ≥ 200 kDa was released from the CHO cells during the production phase. The enzyme expressed collagenase/gelatinase activity when activated. Due to autoprolysis, a number of smaller, less specific MMPs were formed with the smallest form, a 19.4 kDa protein, being the most active. These results may be of particular relevance for other production processes using CHO cells for the expression of recombinant proteins. © 2006 Wiley Periodicals, Inc.

Keywords: proteases; MMP; CHO cells; biotechnology; factor VIII; recombinant proteins

INTRODUCTION

A common cell line used for the production of glycosylated recombinant proteins is the mutant of Chinese hamster ovary (CHO) cells, which lacks the gene for dihydrofolate reductase (DHFR). Its gene amplification can be performed by co-selection with methotrexate, an inhibitor of DHFR, to give rise to high-level gene expression and high protein production. In manufacturing recombinant proteins, the use of serum-free media for cell cultivation is preferred to meet quality requirements. Due to the absence of serum protease inhibitors, however, serum-free production may increase the

exposure of the expressed product to proteolytic attacks that yield protein heterogeneity, denaturation, and loss of protein function. In the present study, we mapped the protease pattern during the production of a model protein, a recombinant factor VIII derivative (r-FVIII) from a CHO (DHFR-) cell line (DG44N.Y) in serum-free medium. Furthermore, we present results from the partial characterization of the proteases that might affect the product integrity, unless controlled.

MATERIALS AND METHODS

Chemicals

Aprotinin, bestatin, E-64, leupeptin, 3,4-dichloroisocoumarin (3,4-DCI), 4-amidinophenylmethanesulfonyl fluoride (PMSF), pepstatin, phosphoramidon, and EDTA were purchased from Boehringer Mannheim Biochemica, Stockholm, Sweden. Cystatin, FA-Gly-Leu-NH₂ (FAGL), FA-Leu-Gly-Pro-Ala (FALGPA), histidine, and imidazol were from Labkemi, Stockholm, Sweden. N-nicotinyl-Tyr-Ne(Na-CBZ-Arg)-Lys-His-Pro-Ile and *p*-amino-phenyl mercuric acetate (APMA) were from Sigma, St. Louis, MO. Pefabloc Try was obtained from Pentapharm Ltd., Basel, Switzerland. Z-Phe-Arg-MCA and Boc-Leu-Arg-MCA were from Peptide Institute, Inc., Osaka, Japan. Antithrombin III and heparin were obtained from Pharmacia, Stockholm, Sweden. Glu-Pro-Val-pNA (S 2484), MeO-Suc-Arg-Pro-Tyr-pNA (S 2586), NCbo-Arg-Gly-Arg-pNA (S 2765), and X-CO-Ile-Glu-OR-Gly-Arg-pNA (S-2222) were purchased from Chromogenix AB, Mölndal, Sweden.

CHO Cell Cultivation Process

The CHO cell cultivation process for producing an r-FVIII derivative of 170 kDa was performed by continuously

Correspondence to: M. Wrangel

H. Sandberg's present address is Octapharma AB, SE-112 75 Stockholm, Sweden.

collecting cell culture supernatant from the CHO cell line, DG44N.Y, from the bioreactor using an external cell separator. The cells were returned to the reactor and the cell culture in the reactor was replenished with fresh medium at the same rate as the removal of supernatant. The supernatant was filtered and submitted to the primary capture step on an ion exchange chromatography column (S-Sepharose from Amersham BioSciences, Uppsala, Sweden) yielding a concentration of the product (the IEX-eluate).

Protease Assays

The broad spectrum protease assay was either based on resorufin-labeled casein (Boehringer Mannheim Biochemica) or succinylated casein (QuantiCleave protease assay kit from Pierce, Rockford, IL). Proteolytic cleavage of resorufin-labeled casein generated peptides that were monitored by fluorescence, measured at 574 nm (absorbance) and 584 nm (emission) by a spectrofluorimeter (F4500, from Hitachi Ltd., Tokyo, Japan). The assay procedure described by the manufacturer was followed. Incubation of substrate with proteases was performed for 24 h and then stopped with trichloroacetic acid. The QuantiCleave assay was performed as described by the manufacturer. The kit contained trinitrobenzenesulfonic acid (TNBSA) that reacts with primary amines exposed after the protease cleavage of peptides. Reaction products are measured at 450 nm. This assay may be performed with microtiter plates but is used only for analyzing proteases in samples that are devoid of free amino acids. Thus, it cannot be used to measure proteases in the cell supernatant. Immunoplates (96-well) were used for fluorescence analysis and a spectrofluorimeter (Fluorolog 2 Spex-Horiba, equipped with a microplate attachment) was used for the measurements.

The fluorimetric assay for collagenase was based on use of Mcc-Pro-Leu-Gly-Pro-D-Lys(Dnp)-OH (Pz peptide) from Calbiochem, Merck Bioscience, Darmstadt, Germany, with a method that was previously described (Barett et al., 1989). Other substrates for proteases were used according to the recommendations from the manufacturers. The equipment was the same as that used for the succinylated casein assay.

A broad spectrum assay for measurement of serine proteases was based on the use of the chromogenic substrate, S-2288. Samples were diluted at least twofold in 0.15 M Tris-HCl, pH 7.4, and 100 μ L were added to microtiter plates followed by the addition of 20 μ L of 10 mM substrate. The plates were kept at 37°C for 24 h if not otherwise indicated after which the absorbance at 405 nm of released *p*-nitroanilide was measured.

Factor VIII Activity (VIII:C)

VIII:C was determined by a chromogenic substrate microtiter method using Coatest factor VIII kit from Chromogenix AB. A factor VIII reference preparation calibrated against the 6th International Standard from the National Institute for Biological Standards and Control (NIBSC), was used as the

house standard. All dilutions of the standard and samples were done with 1% bovine serum albumin in the buffer.

Electrophoresis

Non-reduced SDS-PAGE was performed using precast gels from Novex (San Diego, CA). Zymogram SDS-PAGE, for detecting protease activity, was performed with gels copolymerized with 0.1% casein or gelatin. Precast zymogram casein and gelatin gels containing 12 and 10% polyacrylamide, respectively, were obtained from Novex. Sample volumes applied to the gels were 30 μ L. Running conditions were as recommended by the manufacturer. Molecular mass standard proteins were from Invitrogen (Mark 12, Paisley, UK).

N-Terminal Analysis

Proteins were separated by SDS-PAGE and transferred to polyvinylidene difluoride membranes (PVDF) [Problott from Applied BioSystems, Foster City, CA]. Commassie-stained bands were cut out and submitted to N-terminal amino acid sequence analysis using a Procise protein sequencer (Applied Biosystems).

Chromatography

Size exclusion chromatography (SEC) was performed on a Sephacryl S300 column (length: 75 cm, diameter: 2.6 cm) [Amersham Biosciences]. The buffer was 0.05 M HEPES, 0.3 M NaCl, 5 mM CaCl₂, pH 7.0. Hydrophobic interaction chromatography (HIC) was performed with Butyl Sepharose FF from Amersham BioSciences. The gel was packed in a 15 mL column and equilibrated with the HEPES 0.05 M buffer, pH 7.0, with 5 mM CaCl₂, and 1 M NaCl. Samples were applied in the same buffer except that the NaCl concentration was 1.4 M. Following a washing procedure with the buffer at 1 M NaCl, elution was carried out in a decreasing salt gradient (from 1.0 to 0 M NaCl). An Äkta FPLC system (Amersham BioSciences) was used for chromatography.

RESULTS

Mapping of Proteolytic Activities

A casein-based broad spectrum protease assay was used to determine the protease activity in the cell culture supernatant and the eluate from the primary capture step (IEX-eluate). A significant amount of protease activity was found in the eluate from the primary capture step (IEX-eluate), as measured by the casein-resorufin assay. Figure 1 shows the absorbance at 574 nm from the assay at different dilutions of the IEX-eluate, and the corresponding non-diluted cell supernatant. From the figure, at least a 200-fold increase in the protease activity occurred in the primary capture step due to an increased concentration and/or an activation effect.

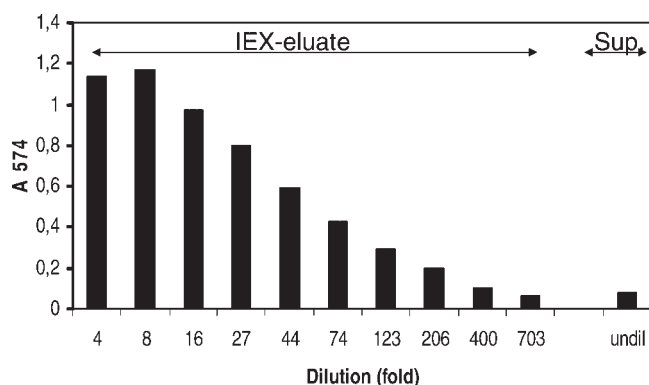


Figure 1. Proteolytic activity in IEX-eluate at different dilutions and in the undiluted cell supernatant. Measurements were made with the broad spectrum protease assay, based on resorufin-labeled casein.

With aim to map the proteolytic activity, experiments were conducted by adding different protease inhibitors to the cell culture supernatant or the IEX eluate. The inhibitors used comprised of metal chelating agents such as EDTA, 1,10 *o*-phenantroline, histidine, and imidazole—all potential metalloproteinase inhibitors; serine protease inhibitors aprotinin, 3,4-DCI, Pefabloc Try, and PMSF; leupeptin—able to inhibit both serine and cysteine proteases; cysteine protease inhibitors—cystatin and E-64; bestatin—an inhibitor of aminopeptidase (an exopeptidase of the metalloproteinase-type); and pepstatin—an inhibitor of aspartic proteases. The inhibitor concentrations used were first the starting concentrations recommended for normal substrate inhibition by the suppliers. However, in most cases higher concentrations of inhibitors were then used if no or only partial inhibition of caseinase activity was achieved. The inhibitors were incubated with samples for 1–3 h prior to analysis with the protease assay.

In Table I, the effects from addition of various inhibitors to the undiluted IEX-eluate are shown, as measured by the casein assay. EDTA at 4 mM and 1,10 *o*-phenantroline at 1 mM concentrations efficiently inhibited the protease

Table I. Effect of general protease inhibitors on caseinase activity in IEX eluate.^a

Protease inhibitor	Concentration (mM)	Inhibition (%)
EDTA	4	96.0 ± 2.8 (n = 2)
1,10 <i>o</i> -phenantroline	1	98.0 ± 2.1 (n = 3)
Aprotinin	0.15	1.0 ± 2.0 (n = 4)
3,4 dichloroisocoumarin	0.3	0 (n = 2)
Pefabloc Try	0.036	0 (n = 2)
Leupeptin	1 × 10 ⁻³	12.7 ± 7.5 (n = 3)
PMSF	1	10.3 ± 8.5 (n = 3)
Cystatin	2 × 10 ⁻⁵	0 (n = 2)
E-64	0.056	3.8 ± 4.4 (n = 4)
Bestatin	1.6	11.5 ± 3.5 (n = 2)
Pepstatin	0.2	16.5 ± 5.0 (n = 2)

^aThe data represent results from separate experiments with different samples. Samples were incubated for 3 h at room temperature in the presence of the inhibitors. The broad spectrum protease assay was based on succinylated casein.

activity (almost 100% inhibition). Although none of the other inhibitors demonstrated comparable effects, a low degree of inhibition, 2–20%, was achieved with the serine protease inhibitors PMSF and leupeptin; the cysteine protease inhibitor, E-64; the aminopeptidase inhibitor, bestatin; and the aspartic protease inhibitor, pepstatin. Thus, the metalloproteinase(s) are assumed to be responsible for the greater part of the protease activity liberated by the CHO cells. A smaller portion of the protease activity was due to the presence of serine and cysteine proteases, which has been described previously in the supernatant of CHO cells (Sato et al., 1990). Further evidence for the presence of cysteine proteases, as measured in the cell supernatant, was seen by the significant cleavage occurring with peptide substrates Z-Phe-Arg-MCA and Boc-Leu-Arg-Arg-MCA, which were previously reported to be sensitive to cleavage by cysteine proteases (Sato et al., 1990). Fluorescence (absorbance at 380 nm and emission at 460 nm), measured after incubation with the two substrates for 22 h at 37°C, showed a 10-fold and a 15-fold increase in fluorescence over the controls. Unexpectedly, some aspartic protease activity seemed to be present as well, in spite of that the test was performed at pH 7.5. Most aspartic proteases are reported to be inactive at a pH above 6.0 and due to their low pH optimum, are deemed to be unlikely sources of activity during cell cultivation and the following protein purification (Salveson and Nagase, 1989).

Evaluation of the Effects of Metal Chelators on Caseinase Activity and Factor VIII Stability

Further evaluations of the inhibitory effects of metal chelators on caseinase activity were conducted. Figure 2 shows that a concentration of more than 1 mM EDTA, added to the cell supernatant, effectively inhibited the caseinase activity. Also the chelating agents histidine and imidazole were found to inhibit the caseinase activity but at a higher molar excess of the chelators. Thus, all chelators had inhibitory effects on the caseinase activity, with the most effective being EDTA and 1,10 *o*-phenantroline, that yielded 95–100% inhibition. Non-chelating, metalloproteinase inhibitors like phosphoramidon (inhibitor of some metalloendopeptidases) (Barett, 1994; Bond and Beynon, 1985) at 0.01 mM, Cpp-Ala-Ala-Phe-pAB (Orlowski et al., 1988) at 0.5 mM and N-nicotinyl-Tyr-Ne(Na-CBZ-Arg)-Lys-His-Pro-Ile (inhibitor of angiotensin II) at 0.3 mM concentrations showed only a low degree, less than 20%, of inhibition. As shown in Figure 3, the stability of factor VIII activity in the IEX-eluate was found to be markedly improved by the addition of metal chelators EDTA or 1,10 *o*-phenantroline. In contrast, diprotin A and B (inhibitors of aminopeptidases of the exopeptidase type) had no influence on the stability. When chelators were added to the IEX-eluate, the factor VIII activity was completely stable for at least 24 h at room temperature. A similar stabilizing effect on the factor VIII activity was also achieved by storing the cell supernatant after adding EDTA. As the addition of the inhibitors to the aminopeptidases had no effect on factor VIII stability

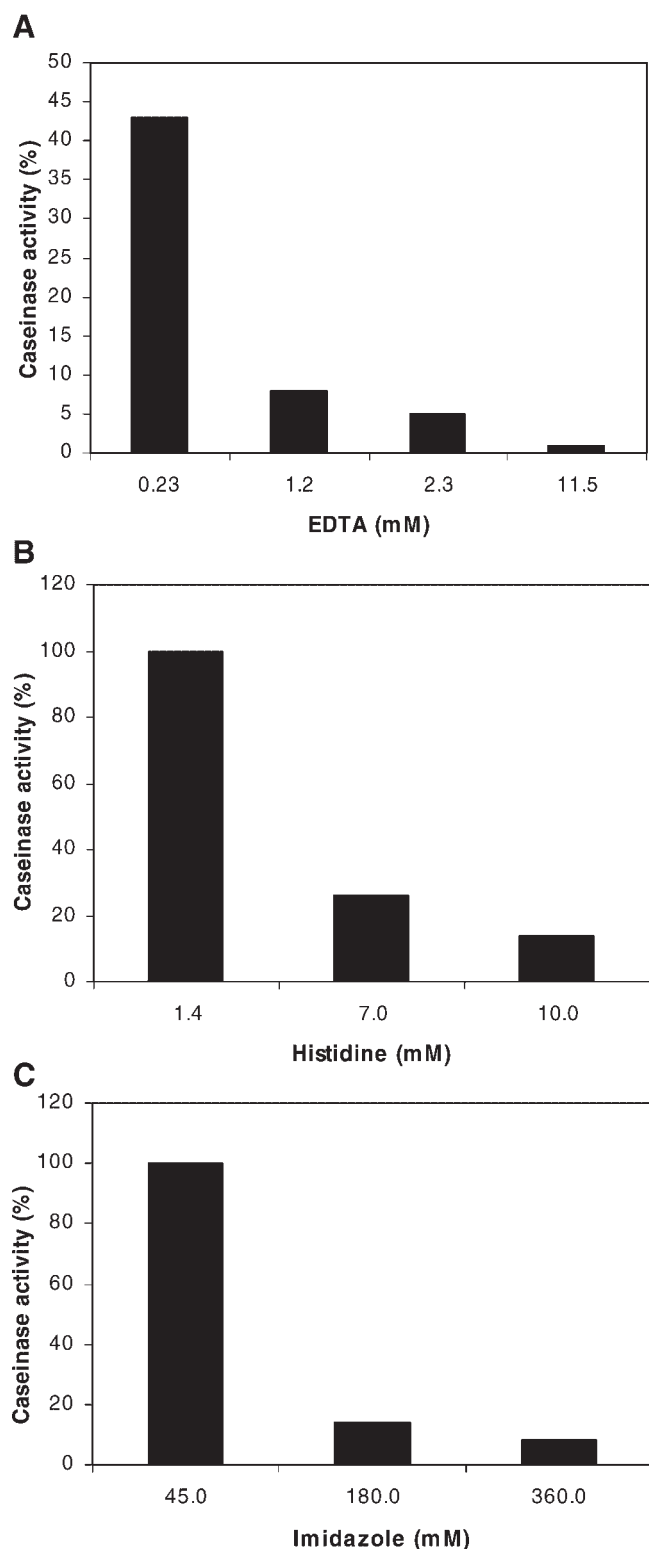


Figure 2. Inhibitory effects on protease activity in the cell supernatant by the chelating agents, EDTA, histidine, and imidazole at different concentrations. Samples were incubated for 2 h at 37°C with the chelating agents. The broad spectrum protease assay, based on resorufin-labeled casein, was used for measurements. The protease activity is expressed as percent (%) of caseinase activity present in the cell supernatant with no added chelating agent.

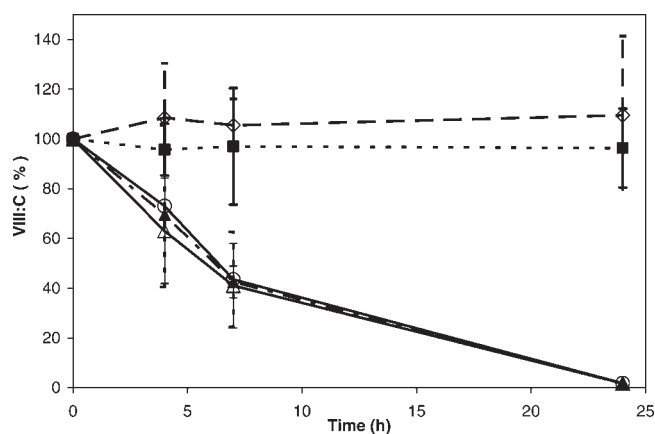


Figure 3. Effect of metalloproteinase inhibitors on factor VIII stability in IEX-eluate. Samples were incubated for 24 h at room temperature. At specific time intervals, samples were taken for measuring factor VIII activity, as indicated. Inhibitors added were: EDTA (2.8 mM), $n = 2$, (◇); *o*-phenantroline (5.5 mM), $n = 3$, (■); diprotin A (68 μM), $n = 2$, (△); and diprotin B (158 μM), $n = 2$, (▲). The sample of IEX-eluate with no added inhibitor is denoted as (○), $n = 2$. The data points represent separate experiments with two or three different sample preparations. Factor VIII activity is expressed as percent (%) of the initial value.

suggested that metalloproteinases of the exopeptidase type were not present. Thus, it seems that the only CHO cell-derived protease activity causing the factor VIII instability in harvest and the IEX-eluate originated from metalloproteinase(s) of the endopeptidase type.

Specificity of the Serine Protease

Although the CHO cell-derived serine protease did not appear to be important for destabilizing the factor VIII during production, further studies of this activity may be warranted for the production of other recombinant proteins using the CHO cells. We, therefore, performed additional assays on the CHO cell-derived serine protease activity in both the IEX-eluate and in the cell supernatant, using the broad spectrum serine protease substrate, S-2288. In this case, the same amount of protease activity was found in both the cell supernatant and in the IEX-eluate, so that no concentration/activation of the protease occurred in the primary capture step. When different serine protease inhibitors were added, similar effects were found (Table II). Aprotinin yielded a low inhibition of activity (15–25%) while 3,4-DCI, Pefabloc Try, leupeptin, and PMSF inhibited the protease activity in the range of 40–100%. The combination of antithrombin III and heparin, known to inhibit vitamin K-dependent clotting enzymes, was one of the most efficient inhibitors of the serine protease activity with a total inhibition attained. The nature of the serine protease activity was further examined using additional chromogenic substrates. These were: Glu-Pro-Val-pNA (S 2484), specific for granulocyte elastase; MeO-Suc-Arg-Pro-Tyr-pNA (S 2586), for monitoring chymotrypsin; NCbo-Arg-Gly-Arg-pNA (S 2765), having high specificity for coagulation factor Xa; and X-CO-Ile-Glu-OR-Gly-Arg-pNA for measuring trypsin and factor Xa (S 2222). No

Table II. Effect of serine protease inhibitors on the protease activity in the IEX eluate and cell supernatant as measured with a broad spectrum assay for serine proteases with use of the chromogenic substrate S-2288.^a

Protease inhibitor	Concentration (mM)	Inhibition in IEX eluate (%)	Inhibition in cell supernatant (%)
Aprotinin	0.15	22.0 ± 5.7 (<i>n</i> = 2)	20.7 ± 6.0 (<i>n</i> = 3)
3,4 dichloroisocoumarin	1	88.0 ± 17.0 (<i>n</i> = 2)	57.3 ± 20.2 (<i>n</i> = 4)
Pefabloc Try	0.036	65.0 (<i>n</i> = 1)	89.3 ± 16.8 (<i>n</i> = 3)
Leupeptin	1 × 10 ⁻³	90.3.0 ± 8.5 (<i>n</i> = 3)	95.0 ± 8.7 (<i>n</i> = 3)
PMSF	1	78.5 ± 10.6 (<i>n</i> = 2)	77.0 ± 7.2 (<i>n</i> = 3)
ATIII, 0.1 U/mL + heparin, 3 U/mL	—	100.0 ± 0 (<i>n</i> = 2)	100.0 ± 0 (<i>n</i> = 2)

^aThe data represent results from separate experiments with different samples. The samples with added inhibitors were incubated for 3 hours at room temperature prior to analysis with the chromogenic substrate, S-2288.

reaction was detected with the chromogenic substrates S 2484 or S 2586 when incubated with the cell supernatant. However, considerable cleavage of the factor Xa-specific substrate S 2765 occurred, while S 2222 was cleaved at a low rate. Figure 4 compares the results for S 2765 and the broad spectrum chromogenic substrate S 2288. The *K_m* values were 0.7 mM for S 2765 and 2 mM for S 2288 as determined by measurement of initial reaction rate velocity. The good reactivity of the serine protease with the factor Xa-specific substrate S 2765 taken together with the inhibitory effect caused by antithrombin III, in combination with heparin, suggests that the serine protease activity of the CHO cells was closely related to factor Xa.

Partial Characterization of the Enzyme(s) Responsible for Metalloproteinase Activity During CHO Cell Cultivation

Several types of tests were used in the following to characterize the protease(s) derived from the CHO cell line during cell cultivation with emphasis on the protease(s) expressing metalloproteinase activity.

Specificity and Mode of Activation of Proteases Expressing Metalloproteinase Activity

To analyze protease specificity, different synthetic substrates for metalloproteinases were tested. The proteases from the IEX-eluate showed reactivity with the motifs of Leu-Gly-Pro-Ala (FALGPA) and Mcc-Pro-Leu-Gly-Pro-D-Lys-OH (Pz-peptide), which are both specific for collagenase (Barett et al., 1989; van Wart and Steinbrink, 1981). No response to the substrate, FA-Gly-Leu-NH₂ (FAGL), specific for thermolysin, was seen.

Another type of analysis used casein zymogram and gelatin zymogram SDS-PAGE. Figure 5A shows the pattern of protease activity obtained in casein zymogram gel for a sample of IEX-eluate that had been preincubated at 37°C for different times and at pH 7.05 and 8.8. The colorless bands correspond to zones without casein, due to digestion by proteases. At the start of incubation and pH 7.05, a colorless band appeared at about 55 kDa. This band gradually increased with time and the 55-kDa band appeared with a stronger intensity during incubation at pH 8.8. Furthermore, lower molecular mass bands were seen at 33 and 20 kDa, and

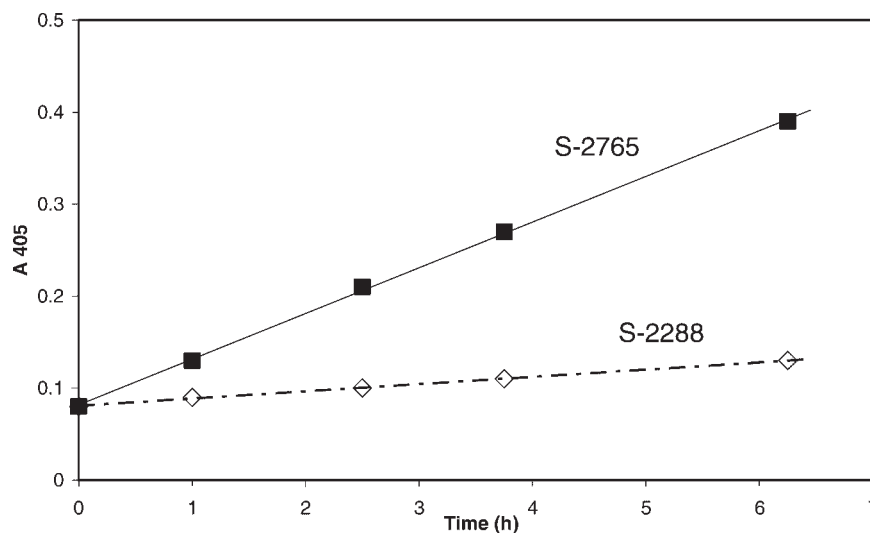


Figure 4. Serine protease activity over time in the cell supernatant using two different chromogenic substrates. The serine protease activity in the cell supernatant was followed for 7 h. Samples were taken at specific time intervals for analysis with the broad spectrum chromogenic substrate for serine proteases, S-2288, and the factor Xa-specific chromogenic substrate, S-2765. Measurements of released *p*-nitroanilide from the substrates were made at 405 nm.

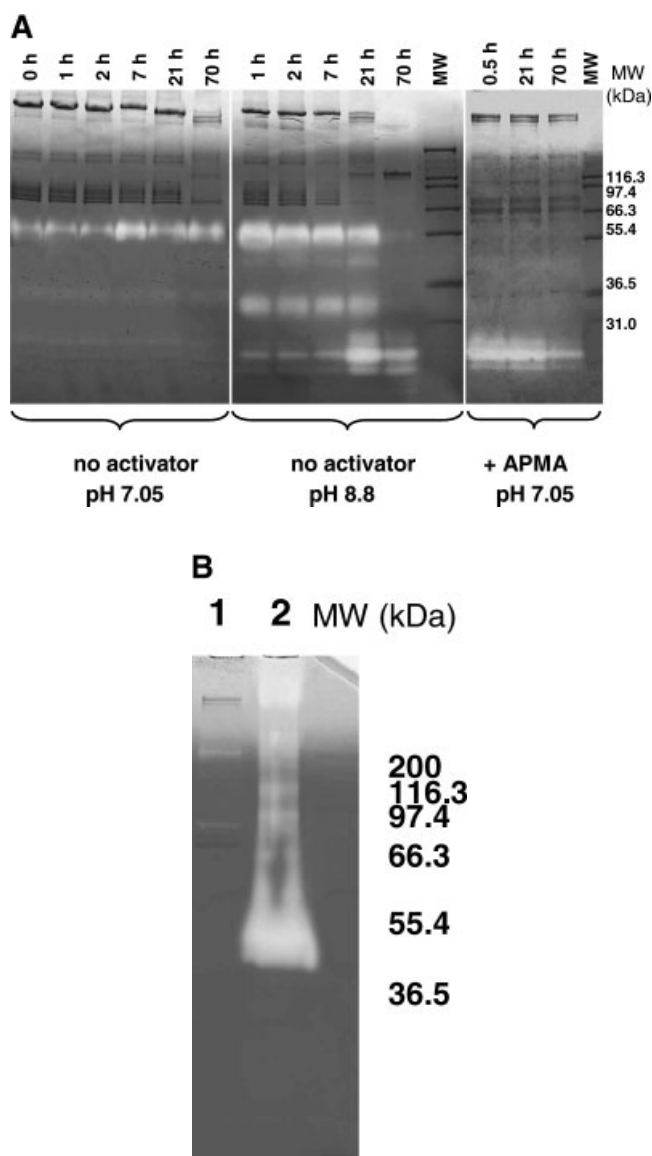


Figure 5. Caseinase and gelatinase activities in the IEX-eluate during incubation, as visualized by Zymogram SDS–PAGE. **A:** Casein, 12% gel, non-reducing. The IEX-eluate, was incubated at 37°C at pH 7.05 and 8.8 for up to 70 h, and at pH 7.05 for up to 70 h in the presence of the MMP-activator, 0.5 mM *p*-amino phenyl mercuric acetate (APMA). Incubation times are specified above the lanes. The optical density of the applied samples was A280 = 1.8. **B:** Gelatin, 10% gel, non-reducing. Lane 1—control, not incubated IEX-eluate, 20-fold concentrated; Lane 2—IEX-eluate, 20-fold concentrated, incubated at 37°C at pH 7.05 for 48 h. The sample buffer was 50 mM Tris-HCl, 5 mM CaCl₂, 0.05% Na₂S₂O₅. Lanes with molecular mass standard proteins were indicated with MW.

these gradually increased during 21 h of incubation and at the higher pH. The 20-kDa band appeared with its strongest intensity after 21 h of incubation. The 33- and 20-kDa species were assumed to have originated from the 55-kDa species. These results indicate that the protease activity increased with time and increasing pH during the test interval, probably due to self-activation, see Discussion. After 70 h, most of the protease activity had faded at pH 8.8. The dominant band was seen at 20 kDa but now stained at a lesser degree. The less intense staining bands, after 70 h, might have been the result

of inactivation of the protease by self-degradation. The increased activity seen in casein zymogram SDS–PAGE at the higher pH also agrees with the results of the casein activity assay. In this case, a linear increase in activity was seen between pH 5 and 9. The protease activity could also increase with *p*-amino phenyl mercuric acetate (APMA), an organo-mercuric compound reported to activate pro-proteins of metalloproteinases (Galazaka et al., 1996). Figure 5A shows the effect of APMA on the protease activity at pH 7.05, as measured by casein zymogen SDS–PAGE. The results were similar to those obtained from incubation at pH 8.8, as both 33- and 20-kDa species appeared. Nevertheless, the 55-kDa species was not detected after addition of APMA, possibly due to its rapid degradation in parallel to generation of the 33- and 20-kDa species. Figure 5B shows the pattern of bands obtained for the 20-fold concentrated IEX-eluate, using gelatin zymogram SDS–PAGE. Two, thin colorless bands appeared at positions corresponding to about 95 and 200 kDa in the run of the non-incubated IEX-eluate whereas intense bands appeared in the molecular mass range from about 40 kDa to >200 kDa with a dominating form appearing at 40 kDa after incubation at 37°C for 2 days at pH 7.0. Thus, activation of the protease(s) had occurred during incubation. These results indicate that the zymogram SDS–PAGE with gelatin detected protease(s) that exhibited a different specificity compared to protease(s) detected by the casein for zymogram SDS–PAGE. Furthermore, the cleavage of gelatin observed here conforms to the positive response seen with the collagenase-specific substrates (FALGPA and Pz-peptide), described above, as gelatin is simply denatured collagen with some loss of the collagen triple helix.

Fractionation of Caseinase, Collagenase, and Serine Protease Activities by Size Exclusion Chromatography (SEC) of IEX Eluate

To further characterize the protease activity that could be neutralized by chelating agents, the IEX-eluate was fractionated by SEC on Sephacryl S 300. The fractions were analyzed by measuring UV absorbance, factor VIII activity, proteinase activity based on the use of succinylated casein, collagenase activity based on the Pz peptide described above, and serine protease activity. Figure 6A and B shows the chromatogram. Factor VIII:C elution was at least partially separated from protease activity suggesting that the factor VIII protein was not tightly associated with any protease. The serine protease activity eluted at a position corresponding to proteins with molecular masses ranging from about 150 to 400 kDa. The high molecular mass might have been caused by an association with another protein or aggregation, since adding 10 mM EDTA to this protease fraction resulted in the formation of a lower molecular mass form. The caseinase activity was found in some SEC fractions at an elution position that corresponded to a protein of molecular mass >600 kDa. The amount of caseinase activity seen at this high molecular mass position varied between runs and in some cases, very low amounts were found. This variation may have

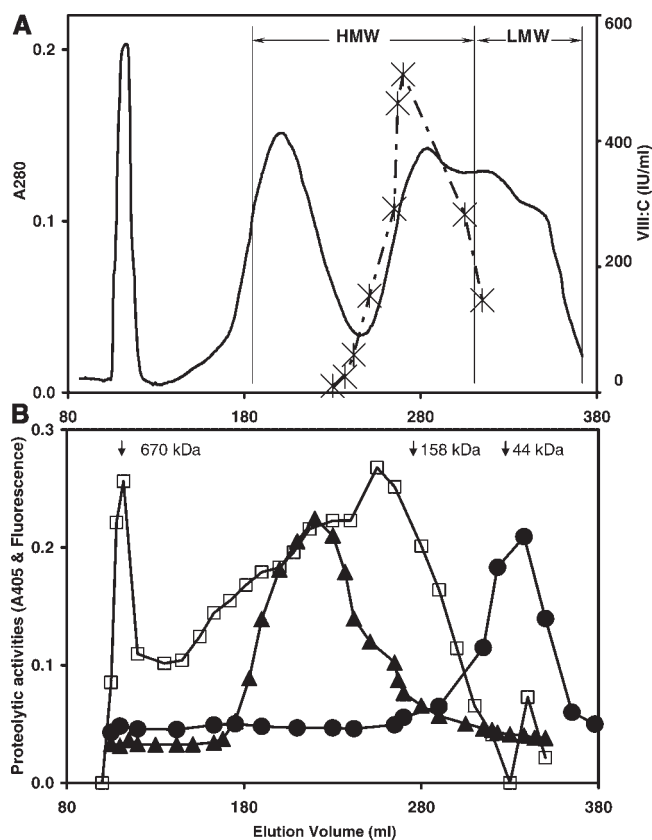


Figure 6. Fractionation of IEX-eluate by SEC on Sephacryl S300. The measured parameters of the fractions from the chromatography are shown in two figures, (A) and (B). The solid line (A) indicates the elution profile measured as optical absorbance at 280 nm. VIII:C (A) denotes factor VIII activity, (*). The protease activities as shown in (B) were: caseinase activity, (●); and collagenase activity measured with the fluorogenic Pz-peptide substrate, (□); both in arbitrary units and serine protease activity at 405 nm with use of the broad spectrum chromogenic substrate for serine proteases, S2288, (▲). HMW and LMW are high- and low-molecular weight pools, respectively, which were subsequently used for purification of metalloproteinases. The elution positions of standard proteins for molecular mass calibration are indicated.

been due to some differences in handling of the samples such as in the time before analysis. Thus, the caseinase activity in this fraction increased with time before the analysis, to suggest that activation of protease had occurred. In contrast, high amounts of caseinase activity was always found at elution positions corresponding to proteins of about 150 kDa to <20 kDa with a maximum at ~20 kDa. Collagenase activity was found in all fractions and showed two peaks with maximum heights at the elution position for proteins of >600 and 200 kDa. The collagenase activity was lowest in the late eluting fractions that contained a high level of caseinase activity.

Partial Characterization of Metalloproteinases Following Hydrophobic Interaction Chromatography of SEC Fractions

The fractions from the gel filtration described above were divided in two pools: one containing the peak fractions at

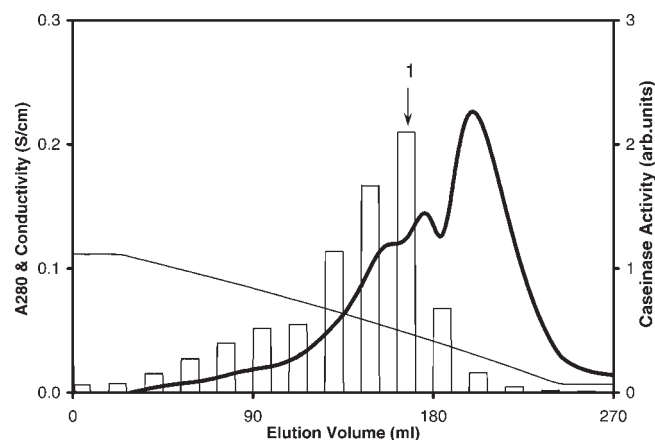


Figure 7. Purification of metalloproteinases by HIC. The low molecular weight pool of SEC fractionation (Fig. 6) was chromatographed on butyl-sepharose. The elution was made in a decreasing salt gradient (thin line) from 1.0 to 0 M NaCl. The elution profile was measured as the optical absorption at 280 nm and is shown by the thick black line. Caseinase activity of selected fractions is shown as bars. The arrow and number 1 denote the fraction that was analyzed by SDS-PAGE and zymogram SDS-PAGE (shown in Fig. 8).

about 20 kDa (LMW) and the other with proteins of higher molecular masses, ~100–300 kDa (HMW). These were both submitted to further purification by HIC on a Butyl-Sepharose FF column. Figure 7 depicts the chromatography of the LMW pool. Silver-stained non-reducing SDS-PAGE of the peak fractions with caseinase activity from this purification step showed that a high degree of purification had been attained. The SDS-PAGE result, after silver-staining (Fig. 8), showed one major band at about 20 kDa and a fainter band right above this. The same sample showed only one band at about 20 kDa on zymogen casein SDS-PAGE (Fig. 8) that closely corresponds to the silver-stained band at 20 kDa. MALDI-MS of the protein showed a double peak with a mean molecular mass of 19.4 kDa. The purified protein was also added to purified factor VIII to study its effect on factor VIII activity. A rapid concentration-dependent decrease of activity was seen in comparison to the control (Fig. 9). Thus, the isolated protease activity was as detrimental to factor VIII as previously seen with the crude proteins in cell supernatant and IEX-eluate.

Figure 10 shows the pattern of bands obtained after submitting the purified LMW (19.4 kDa) and HMW proteins from HIC, to non-reducing SDS-PAGE followed by transfer of bands to a PVDF membrane and Coomassie blue staining for N-terminal amino acid sequence analysis. The same sample, representing the LMW protein described above but loaded in a higher protein amount, was used. The LMW protein appeared as a doublet in this SDS-PAGE, as was also recognized by MALDI-MS. The fainter band immediately above the doublet was identified as a contaminant. Both bands in the LMW doublet had N-terminal amino acid sequences LRELPGMQKW TKHHL TYRIY. A BLAST database search for sequence similarity with other proteins (Altschul et al., 1997) showed between 66 and 80% homology for this sequence with a sequence of the catalytic

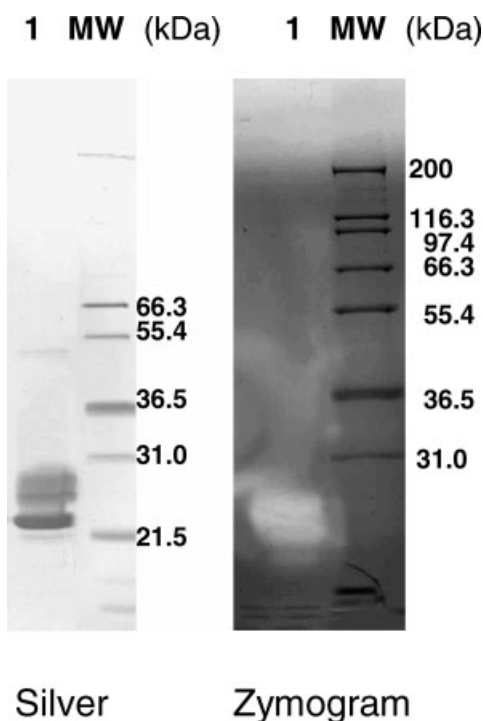


Figure 8. SDS-PAGE and Zymogram casein SDS-PAGE of the metalloproteinase in HIC fraction 1. Left panel—12% non-reducing SDS-PAGE with silver staining. The HIC fraction 1, indicated in Figure 7, with an optical density of $A_{280} = 0.2$, was diluted 1/10 before application to the gel. Right panel—12% Zymogram casein SDS-PAGE of the same fraction, diluted 1/20. Lanes with molecular mass standard proteins were denoted MW.

domain of Stromelysin-1 and Stromelysin-2 that are matrix metalloproteinases (MMPs) 3 and 10 (MMP3 and MMP10), respectively, from man, horse, and rabbit. Furthermore, 61% homology was found with the corresponding region of the catalytic domain of human macrophage metalloelastase

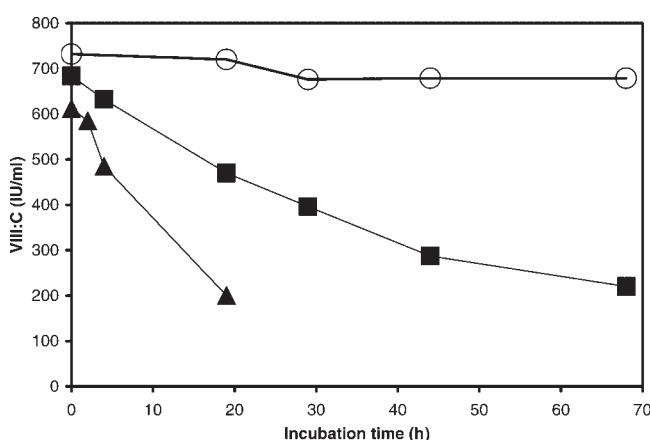


Figure 9. Effect of purified metalloproteinase on factor VIII activity. Control without added metalloproteinase, (○); r-FVIII with metalloproteinase from fraction 1, indicated in Figure 7, at $\sim 0.45 \mu\text{g/mL}$, (■); r-FVIII with metalloproteinase from fraction 1 in Figure 7 at $\sim 0.9 \mu\text{g/mL}$, (▲). Incubation was performed at 22°C in HEPES, 0.05 M buffer, pH 7.2 with 0.3 M NaCl, 9 mM sucrose, 3.4 mM CaCl_2 , and 0.01% Tween 20.

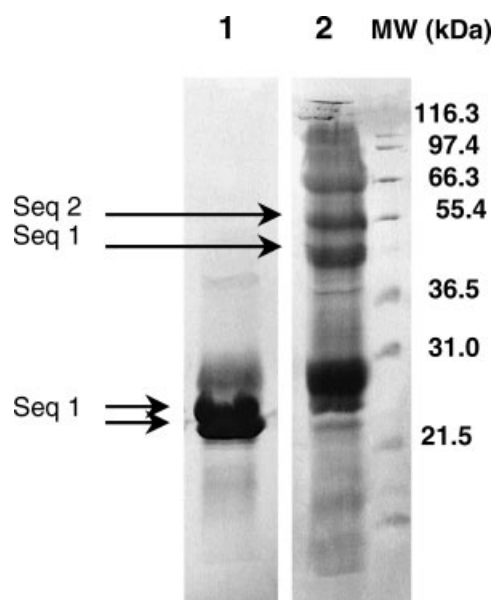


Figure 10. SDS-PAGE of metalloproteinase from HIC fractions for N-terminal amino acid sequencing. Lane 1—LMW fraction 1 of HIC chromatography in Figure 7. Lane 2—HMW fraction of HIC chromatography (similar to that shown in Fig. 7). Samples were concentrated to $\sim 2.5 \text{ mg/mL}$ of protein, applied onto 12% non-reducing SDS-PAGE and blotted onto PVDF membranes and stained with coomassie blue. Sequence 1 denotes the N-terminal sequence LRELPGMQKW TKHHL TYRIY, and sequence 2 was IPI-E TEFAE RYLTK FYDLK. Lane with molecular mass standard proteins was denoted MW.

precursor (MMP12). Table III shows the amino acid alignment between the N-terminal sequence of the LMW protein and the homologous peptides in the data base. The SDS-PAGE of the purified HMW fraction showed a pattern of several bands (Fig. 10). A major portion of the material obviously contained proteins of lower molecular masses than indicated by gel filtration, which might have been caused either by the denaturing electrophoresis conditions that would dissociate possible non-covalently associated protein complexes or the degradation of proteins that might have occurred during the further purification. Subsequent N-terminal sequencing revealed a lower purity in this fraction compared to the LMW protein. Bands appearing at 40 and 55 kDa were found to contain proteins with the N-terminal amino acid sequences of LRELPGMQKW TKHHL TYRIY and IPI-E TEFAE RYLTK FYDLK, respectively. The first sequence was the same as that of the 19.4-kDa protein. The BLAST database search showed that the second sequence had sequence homology (64%) with the propeptides of rat macrophage metalloelastase, MMP12 and mouse MMP12, (Table IV). The catalytic domain of the latter proteinases also had some sequence homology, 55%, with the N-terminal sequence of the isolated 19.4/40-kDa proteins (not shown).

From these results, strong evidence suggests that the 19.4- and 40-kDa proteins were of the same origin. It is quite possible that the N-terminal sequenced 40-kDa species was identical to the intense 40-kDa band seen on gelatin zymogen SDS-PAGE, after incubation of the IEX-eluate at 37°C .

Table III. Search for coincidence of the sequence 1: LRELPGMQKW TKHHL TYRIY and similar sequences detected by the Blast^a search.

Number	Protein from GenBank		Identity (%)	Query subject
	GenBank accession number	Protein definition		
1	AAB05774.1	Matrix metalloproteinase 3 [<i>Equus caballus</i>]	80	PGMQKWTKHHLTYRI PGM KW+K HLTYRI PGMPKWSKTHLTTRI
2	AAB23547.1	Prostromelysin {N-terminal} [dogs, synovial fibroblasts, Peptide Partial, 24 aa, segment 2 of 2]	80	PGMQKWTKHHLTYRI PGM KW K HLTYRI PGMPKWRKXHLTYRI
3	S23518	Stromelysin (EC 3.4.24.-) dog (fragment)	80	PGMQKWTKHHLTYRI PGM KW K HLTYRI PGMPKWRKXHLTYRI
4	BAD96321.1	Matrix metalloproteinase 10 preproprotein variant [Homo sapiens]	80	PGMQKWTKHHLTYRI PGM KW K HLTYRI PGMPKWRKTHLTTRI
5	MMP10_HUMAN	Stromelysin-2 precursor (Matrix metalloproteinase-10) (MMP-10)	80	PGMQKWTKHHLTYRI PGM KW K HLTYRI PGMPKWRKTHLTTRI
6	AAH74815.1	Matrix metalloproteinase 3, preproprotein [Homo sapiens]	66	RELPGMQKWTKHHLTYRI R PG+ KW K HLTYRI RTFPGIPKWRKTHLTTRI
7	MMP3_HUMAN	Stromelysin-1 precursor (Matrix metalloproteinase-3) (MMP-3)	66	RELPGMQKWTKHHLTYRI R PG+ KW K HLTYRI RTFPGIPKWRKTHLTTRI
8	MMP3_RABBIT	Stromelysin-1 precursor (Matrix metalloproteinase-3) (MMP-3)	80	PGMQKWTKHHLTYRI PG KWT K HLTYRI PGTPKWTKTHLTTRI
9	MMP12_HUMAN	Macrophage metalloelastase precursor (EC 3.4.24.65) (HME) (Matrix metalloproteinase-12)(MMP-12) (macrophage elastase) (ME)	61	RELPGMQKWTKHHLTYRI RE+PG W KH++TYRI REMPGGPVWRKHYYTYRI

^aThe sequence search was performed with the help of the BLASTP 2.2.12 program (Altschul et al., 1997). For definition of BLAST parameters, see Altschul et al. (1997). Eight sequence hits of maximal similarity were selected. In addition, the last, ninth row shows the similarity with a MMP12 sequence.

Table IV. Results of search of coincidence for the sequence 2: IPI-E TEFAE RYLTK FYDLK and similar sequences detected by the Blast^a search.

Number	Protein from GenBank		Identity (%)	Query subject
	GenBank accession number	Protein definition		
1	P34960	Macrophage metalloelastase precursor (MME) (Matrix metalloproteinase-12)	64	ETEF AERYLTKFYD ++EFAE YL++FYD DSEFAEWYLSRFYD
2	NP_032631.1 AAA39526.1	Matrix metalloproteinase 12 [<i>Mus musculus</i>]	64	ETEF AERYLTKFYD ++EFAE YL++FYD DSEFAEWYLSRFYD
3	CAJ18584.1 AAH19135.1	Mmp12 [<i>Mus musculus</i>]	64	ETEF AERYLTKFYD ++EFAE YL++FYD DSEFAEWYLSRFYD
4	NP_446415.1	Matrix metalloproteinase 12 [<i>Rattus norvegicus</i>]	64	ETEF AERYLTKFYD E+EFAE YL++F+D ESEFAEWYLSRFFD
5	CAA67142.1	Macrophage metalloelastase precursor [<i>Rattus norvegicus</i>]	64	ETEF AERYLTKFYD E+EFAE YL++F+D ESEFAEWYLSRFFD
6	MMP12_RAT	Macrophage metalloelastase precursor (MME) (Matrix metalloproteinase-12)	64	ETEF AERYLTKFYD E+EFAE YL++F+D ESEFAEWYLSRFFD

^aThe sequence search was performed with the help of the BLASTP 2.2.12 program (Altschul et al., 1997). For definition of BLAST parameters, see Altschul et al. (1997). Six sequence hits of maximal similarity were selected.

Furthermore, the purified 19.4-kDa protein was obviously identical to the potent species seen at 20 kDa in the zymogram casein SDS–PAGE of the IEX-eluate, which seemingly originated from the 33- and 55-kDa protein species, appearing after activation with APMA or by high pH. Also, we propose that the 55-kDa species seen in casein zymograms was identical to the N-terminal sequenced 55-kDa protein.

In summary, the results suggest that all protease species at 20 (19.4), 33, 40, and 55 kDa were of the same origin.

DISCUSSION

The present study describes the mapping of proteolytic activities released by CHO cells in a cell cultivation process and the partial characterization of two types of proteolytic activities: (1) originating from metalloproteinases and (2) from a serine protease. The serine protease activity was due to a factor Xa-like serine protease that was found using a factor Xa-specific chromogenic substrate. Antithrombin III, a potent inhibitor of the clotting enzyme factor Xa and other Vitamin K-dependent clotting enzymes, when combined with heparin, efficiently inhibited the serine protease activity of the CHO cells. To our knowledge, this type of serine protease expression in CHO cells has not been previously described. The presence of this serine protease activity did not seem to have any negative effect on the stability of factor VIII, the model protein chosen for production in this study.

In contrast to the serine protease activity described, metalloproteinases have been found to have a strongly negative effect on the model protein. Chelating agents efficiently blocked the proteinase activity as well as the degradation of factor VIII activity in both the supernatant and in the IEX-eluate. Metalloproteinases are classified on the basis of a presence of a metal ion (usually zinc) that participates in the catalysis. The metalloproteinases are differentiated from proteinases whose activities are stabilized by the presence of calcium. The delineation of these proteases as being metalloproteinases and not metallo-dependent proteases was based on several findings. An important observation was that the chelator, 1,10 *o*-phenantroline, blocked the proteinase activity. This inhibitor has a much higher binding constant for zinc than for calcium and will not remove calcium from a calcium-binding protein. Therefore, this chelator is diagnostic for a zinc metalloproteinase (Salveson and Nagase, 1989). Furthermore, the proteinases reported here were activated by APMA, an organomercuric compound reported to activate matrix metalloproteinases (MMPs) through dissociating the single cysteine residue in the propeptide from the active site zinc (Bond and Beynon, 1985). Finally, the results obtained from the N-terminal sequence determination of protein bands from the SDS–PAGE gels run on purified proteinase fractions showed sequences that were homologous with sequences present in MMP3, MMP10, and MMP12. To our knowledge, MMPs from CHO cells have been described only in two

studies and were assigned to MMP9 (Elliott et al., 2003; Mols et al., 2001).

Mammalian metalloproteinases range in molecular mass from 17 to 800 kDa (Bond and Butler, 1987). To some extent, these values should be viewed cautiously as some of the enzymes tend to aggregate. The nine classes of MMPs have similar domain structures consisting of a propeptide, a catalytic domain containing Zn^{2+} and Ca^{2+} , and a pexin-like region. All have a high degree of sequence homology. Virtually, all cleave gelatin and most cleave type IV and V collagens (Birkedal-Hansen et al., 1993). Physiologic activation of MMPs is probably initiated by proteases that cleave specific sites within the propeptide, but final processing to the mature form (the active MMP), that lacks the entire propeptide, often requires intermolecular, autoproteolytic cleavage by the target MMP (Morgunova et al., 1999). When triggered by sulfhydryl-reactive compounds like APMA, the processing of proMMPs to the active form can be entirely autoproteolytic. The C-terminal hemopexin-like domain is linked to the catalytic domain and may determine substrate specificity. As autoproteolysis proceeds and smaller active forms are generated, the enzyme loses much of its specificity and acts as a non-specific enzyme.

Several proteinase species with caseinase or collagenase/gelatinase activity were found in the present study. In the casein zymogram SDS–PAGE, one species at 55 kDa had caseinase activity. Further activation generated new species with activity at about 33 and 20 kDa that was in parallel to the disappearance of the 55-kDa band, suggesting a relationship between these species. Furthermore, in gelatin (denatured collagen) zymograms, activity was found for the species at 95 and 200 kDa, and following activation, at 40 kDa, as well as a range of higher molecular mass proteases, up to above 200 kDa. SEC revealed a further complex pattern. The major part of the caseinase activity was eluted as a peak at a position corresponding to a molecular mass range of <20 kDa to ~150 kDa with a maximum at 20 kDa. High amounts of collagenase activity was found at the high molecular mass position >600 kDa while often rather small amounts of caseinase activity was seen here. The major part of collagenase activity, however, was found as a broad peak with its maximum at the elution position of 200 kDa. In contrast, the low molecular mass form of proteases, cleaved the collagenase-specific substrate (the Pz-peptide), at a lower rate, compared to the cleavage of casein. This suggests that a high molecular mass collagenase was liberated from the CHO cells during production. This was confirmed by the positive reaction with another collagenase substrate, FALGPA. This observation, together with the fact that gelatin was digested in the zymogram SDS–PAGE, indicated the presence of a collagenase.

The purification and subsequent analysis of the proteases showed that the species at ~20 kDa with caseinase activity had a molecular mass of 19.4 kDa and a high level of metalloproteinase activity, as judged by zymogen SDS–PAGE with casein, and the potential to inactivate factor VIII. The 40-kDa protein that exhibited collagenase activity had

the same N-terminal amino acid sequence as the casein-sensitive, 19.4-kDa protein, while the 55-kDa protein that had caseinase activity had a different N-terminal sequence.

The BLAST data base search made showed homologies for the N-terminal amino acid sequence of both the 19.4/40-kDa proteins and the 55-kDa protein with sequences present in MMP 3, MMP10, and MMP12 although the best homology for the 19.4/40-kDa proteins were with sequences present in the catalytic domain of MMP 3/MMP10 and for the 55-kDa protein with a sequence in the propeptide of MMP12. However, the comparison of sequences made suffers from the lack of hamster-specific metalloproteinase in the data base. Furthermore, only limited amino acid sequences from the purified fractions were available for the data base search. This might explain the somewhat discrepant results obtained for the 19.4/40- and the 55-kDa proteins in the BLAST data base search. Therefore, there is no conclusive data here as regards the MMP class of the purified proteinase fractions. The findings from this study, however, when combined with the general knowledge about the nature of MMPs and the amino acid sequence homology between different MMPs, create the following picture: It is likely that the 55-kDa protein contained the propeptide of an MMP as its N-terminal amino acid sequence had certain homology with a sequence found in the propeptide of MMP12 while the lower molecular mass forms, 19.4- and 40-kDa proteins, both having an N-terminal sequence in homology with that of the conserved catalytic region of MMPs 3 and 10, from several species and to a lesser degree with that of MMP 12, lacked the propeptide, and thus would represent more active forms than the 55-kDa protein. Although no amino acid sequence was obtained for the 33-kDa protein, it is likely from the experiments performed that it originated from the 55-kDa protein and was a more active form than the latter. The interpretation of the data was that the 19.4, 33, 40, and 55 kDa forms of metalloproteinase species were all of the same origin. Based on the SEC data and the gelatin zymogram SDS-PAGE, these different molecular mass species were probably derived from the higher molecular mass forms. The high rate of cleavage of the casein substrate by the low molecular form proteinases suggests that these forms have different specificities and were acting more like non-specific enzymes, compared to the high molecular mass forms with their specificity against collagen. This changing pattern of specificity is in line with that reported for MMPs upon autoproteolysis. The spectrum of molecular mass forms of proteinase obtained here is likely due to autoproteolysis that is also characteristic for MMPs. Due to the high degree of autoproteolysis, which, at a late stage results in inactivation of the enzyme, we met with difficulties in calculating the purification folds and specific activities during the purification steps. The high degree of

variation in values was possibly due to differences in the conditions, such as time points, temperatures, etc.

From the data, we propose that an MMP pro-enzyme of about 200 kDa is released from the CHO cells during the production phase for factor VIII. The range of molecular masses, from 200 to >600 kDa, could be attributed to varying degrees of aggregate formation or to associations with other proteins. The enzyme expresses collagenase/gelatinase activity when activated, possibly due to limited autoproteolysis. Further autoproteolysis yields smaller MMPs, which more closely resemble non-specific proteinases.

The results from this study have not been described previously and may be of relevance for other production processes using CHO cells for expression of recombinant proteins.

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